Sangdon Park

Assistant Professor, POSTECH GSAI/CSE

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Google Scholar

Research Interests

My research focuses on the foundation of trustworthy machine learning for safety-critical real-world AI, including agentic and physical AI by developing principled learning algorithms for trustworthy adaptation in the real world.

Education

University of Pennsylvania — Philadelphia, USA
Ph.D. in Computer and Information Science — 2021

- Advisors: Insup Lee and Osbert Bastani
- Thesis: *Uncertainty Estimation Toward Safe AI*
- Committee: Kostas Daniilidis, Nikolai Matni, Edgar Dobriban, and Kilian Q. Weinberger

Seoul National University — Seoul, Korea
M.S. in Electrical and Computer Engineering — 2012

- Advisor: Kyoung Mu Lee
- Thesis: *Abnormal Object Detection by Transformed-Canonical Scene Generation*

Seoul National University — Seoul, Korea
B.S. in Computer Science and Engineering — 2010

- Thesis Advisor: Byoung-Tak Zhang
- Thesis: *Behavioral Intelligence for Crowd Avatar in 3D Virtual Worlds*

Employment

POSTECH — Pohang, Korea
Assistant Professor — Aug. 2023–Now

Georgia Institute of Technology — Atlanta, USA
Postdoctoral Researcher (Mentor: Taesoo Kim) — Sep. 2021–Jul. 2023

Google Cloud AI — Sunnyvale, USA
Research Intern (Host: Kihyuk Sohn) — Summer 2020

Biointelligence Laboratory, Seoul National University — Seoul, Korea
Undergraduate Researcher — 2008–2010

Republic of Korea Army — Korea
Military Service — 2006–2008

Publications

[1] M. Lee, K. Cho, and **S. Park**, *Which Preferences to Train On? End-to-End Multi-Objective Alignment with an Adversarial Preference Distribution*, 2026. arXiv: 2610.04845 [cs.LG].

[2] M. Lee*, Y. Jung*, and **S. Park**, "Online Conformal Abstention Under Adversarial Bandit Feedback", in The Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026.

[3] J. Park, N. Park, S. Lee, Y.-C. Jhi, J. Cho, and **S. Park**, "MJ: Multi-turn LLM Jailbreaking via Decomposed Credit Assignment", in The Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026.

[4] M. Park*, S. Lee*, K. Kim, J. Heo, S. Cho, S. Li, **S. Park**, and D. Kim, "Transductive Generalization for GNNs via Optimal Transport", in The Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026.

[5] J. Choi*, J. Yang*, and **S. Park**, "SAUL: Sharpness-Aware Augmented-Lagrangian Unlearning", in The 2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP Findings**), 2026.

[6] K. Kim*, Y. Choi*, S. Lee, S. Jun, D. Kim†, and **S. Park**†, "The Interplay of Harness Design and Post-Training in LLM Agents", in The 2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP Findings**), 2026.

[7] S. Lee, H. Kim, and **S. Park**, "Can We Stop Malicious AI? KILLBENCH: A Benchmark for External AI Kill Switch Feasibility", in The 2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP Findings**), 2026.

[8] X. Cheng, F. Sang, Y. Yang, H. Zhang, **S. Park**, X. Zhang, and T. Kim, "The Illusion of Rust Safety: Detecting Modular Unsafe Functions with LLMs", in The ACM Conference on Computer and Communications Security (**CCS**), 2026.

[9] J. Jeong, T. Kim, and **S. Park**, "Towards Functional Correctness of Code Models with Selective Generation", in International Conference on Machine Learning (**ICML**), 2026.

[10] J. Hwang, **S. Park**, and J. Ok, "LLM Watermark Evasion via Bias Inversion", in International Conference on Machine Learning (**ICML**), 2026.

[11] X. Cheng*, **S. Park***, H. Han, X. Zhang, and T. Kim, "Ruby: Unmasking Unsafe Rust in Stripped Binaries via Machine Learning", in International Conference on Dependable Systems and Networks (**DSN**), 2026.

[12] J. Yang*, K. Kim*, and **S. Park**, "Online Conformal Prediction with Adversarial Semi-bandit Feedback via Regret Minimization", in International Conference on Learning Representations (**ICLR**), 2026.

[13] T. Kim, H. Han, S. Park, D. R. Jeong, D. Kim, D. Kim, E. Kim, J. Kim, J. Wang, K. Kim, S. Ji, W. Song, H. Zhao, A. Chin, G. Lee, K. Stevens, M. Alharthi, Y. Zhai, C. Zhang, J. Jang, Y. Jang, A. Askar, D. Kim, F. Fleischer, J. Cho, J. Kim, K. Ko, I. Yun, **S. Park**, D. Baik, H. Lee, H. Heo, M. Gwon, M. Lee, M. Baek, S. Min, W. Kim, Y. Jin, Y. Park, Y. Choi, J. Jung, G. Lee, J. Jang, K. Kim, Y. Cha, and Y. Kim, *ATLANTIS: AI-driven Threat Localization, Analysis, and Triage Intelligence System*, 2025. arXiv: 2509.14589 [cs.CR].

[14] K. Kim, Y. Choi, H. Kim, D. Kim, and **S. Park**, "ChronoBias: A Benchmark for Evaluating Time-conditional Group Bias in the Time-sensitive Knowledge of Large Language Models", in Findings of the Association for Computational Linguistics: EMNLP (**EMNLP Findings**), 2025.

[15] J. Hwang, J. Park, H. Park, D. Kim, **S. Park**, and J. Ok, "Retrieval-Augmented Generation with Estimation of Source Reliability", in Empirical Methods in Natural Language Processing (**EMNLP**), 2025.

[16] S. Moon*, M. Lee*, **S. Park**, and D. Kim, "Holistic Unlearning Benchmark: A Multi-Faceted Evaluation for Text-to-Image Diffusion Model Unlearning", in International Conference on Computer Vision (**ICCV**), 2025.

[17] H. Wang, Z. Yang, **S. Park**, Y. Yang, S. Kim, W. Lunardi, M. Andreoni, T. Kim, and W. Lee, "SOUNDBOOST: Effective RCA and Attack Detection for UAV via Acoustic Side-Channel", in International Conference on Dependable Systems and Networks (**DSN**), 2025.

[18] M. Lee*, K. Kim*, T. Kim, and **S. Park**, "Selective Generation for Controllable Language Models", in Neural Information Processing Systems (**NeurIPS**), 2024.

[19] S. Li, **S. Park**, I. Lee, and O. Bastani, "TRAQ: Trustworthy Retrieval Augmented Question Answering via Conformal Prediction", in North American Chapter of the Association for Computational Linguistics (**NAACL**), 2024.

[20] H. Park*, J. Hwang*, S. Mun, **S. Park**, and J. Ok, "MedBN: Robust Test Time Adaptation against Malicious Test Samples", in Computer Vision and Pattern Recognition (**CVPR**), 2024.

[21] W. Si, **S. Park**, I. Lee, E. Dobriban, and O. Bastani, "PAC Prediction Sets Under Label Shift", in International Conference on Learning Representations (**ICLR**), 2024.

[22] W. Si, S. Li, **S. Park**, I. Lee, and O. Bastani, "Angelic Patches for Improving Third-Party Object Detector Performance", in Computer Vision and Pattern Recognition (**CVPR**), 2023.

[23] **S. Park**, O. Bastani, and T. Kim, "ACon^2: Adaptive Conformal Consensus for Provable Blockchain Oracles", in USENIX Security Symposium (**Security**), 2023.

[24] R. Kaur, K. Sridhar, **S. Park**, Y. Yang, S. Jha, A. Roy, O. Sokolsky, and I. Lee, "CODiT: Conformal Out-of-Distribution Detection in Time-Series Data for Cyber-Physical Systems", in International Conference on Cyber-Physical Systems (**ICCPS**), 2023.

[25] **S. Park**, E. Dobriban, I. Lee, and O. Bastani, "PAC Prediction Sets for Meta-Learning", in Neural Information Processing Systems (**NeurIPS**), 2022.

[26] S. Li, **S. Park**, X. Ji, I. Lee, and O. Bastani, *Towards PAC Multi-Object Detection and Tracking*, 2022. arXiv: 2204.07482 [cs.CV].

[27] S. Jang, **S. Park**, I. Lee, and O. Bastani, "Sequential Covariate Shift Detection Using Classifier Two-Sample Tests", in International Conference on Machine Learning (**ICML**), 2022.

[28] **S. Park**, E. Dobriban, I. Lee, and O. Bastani, "PAC Prediction Sets Under Covariate Shift", in International Conference on Learning Representations (**ICLR**), 2022.

[29] R. Kaur, S. Jha, A. Roy, **S. Park**, E. Dobriban, O. Sokolsky, and I. Lee, "iDECODe: In-distribution Equivariance for Conformal Out-of-distribution Detection", in Association for the Advancement of Artificial Intelligence (**AAAI**), 2021.

[30] **S. Park**, S. Li, I. Lee, and O. Bastani, "PAC Confidence Predictions for Deep Neural Network Classifiers", in International Conference on Learning Representations (**ICLR**), 2021.

[31] **S. Park**, O. Bastani, J. Weimer, and I. Lee, "Calibrated prediction with covariate shift via unsupervised domain adaptation", in International Conference on Artificial Intelligence and Statistics (**AISTATS**), 2020.

[32] **S. Park**, O. Bastani, N. Matni, and I. Lee, "PAC Confidence Sets for Deep Neural Networks via Calibrated Prediction", in International Conference on Learning Representations (**ICLR**), 2020.

[33] **S. Park**, J. Weimer, and I. Lee, "Resilient linear classification: an approach to deal with attacks on training data", in International Conference on Cyber-Physical Systems (**ICCPS**), 2017.

[34] J. Oh, T. M. Howard, M. R. Walter, D. Barber, M. Zhu, **S. Park**, A. Suppe, L. Navarro-Serment, F. Duvallet, A. Boularias, et al., "Integrated intelligence for human-robot teams", in International Symposium on Experimental Robotics (**ISER**), 2016.

[35] **S. Park**, W. Kim, and K. M. Lee, "Abnormal object detection by canonical scene-based contextual model", in European Conference on Computer Vision (**ECCV**), 2012.

Scholarships and Awards

- Best Paper Award from CKAIA (한국인공지능학회) 2026
- DARPA AIxCC Winner as Team Atlanta ($4M Team Prize) 2025
- Best Paper Finalist from CKAIA (한국인공지능학회) 2025
- Bang Seung-Yang Graduate Fellowship from POSTECH CSE (awardee: Jaewoo Jeong) 2025
- POSTECH GSAI BK21 Best Paper Award 2025
- NeurIPS'24 Spotlight Paper (top 2.08%) 2024
- DARPA AIxCC Finalist as Team Atlanta ($2M Team Prize) 2024
- NeurIPS'23 Outstanding Reviewer Award 2023
- ICML'23 TEACH Workshop Best Paper Award 2023
- ICCPS'23 Best Paper Award Finalist 2023
- NeurIPS'21 Outstanding Reviewer Award (top 8% of reviewers) 2021
- Korea Cyber-Security Paper Competition Best Paper Award ($4,500) 2018
- PhD fellowship at University of Pennsylvania 2013–2021
- Distinguished MS Dissertation Award at Seoul National University 2012
- Academic Performance Scholarship 2009
- National Science and Engineering Undergraduate Scholarship 2003– 2008

Service

- **Area Chair**
  NeurIPS'24–26, ICML'25–26, ICLR'26–27
- **Reviewer**
  NeurIPS'21–23, ICML'21–23, ICLR'22–24, AAAI'25, Journal of the Royal Statistical Society: Series B
- **External Reviewer**
  S&P'21, S&P'22, Security'22, Security'23, Security'24, NDSS'24

Teaching

- **Instructor** at POSTECH Fall 2026
  ML (CSED515)
- **Instructor** at POSTECH Spring 2026
  AI Security (CSED490H)
- **Instructor** at POSTECH Fall 2025
  ML (CSED515)
- **Instructor** at POSTECH Spring 2025
  Trustworthy ML (AIGS703L / CSED703L)
- **Instructor** at POSTECH Fall 2024
  Trustworthy ML (AIGS703L / CSED703L)
- **Instructor** at POSTECH Spring 2024
  Discrete Mathematics (CSED261)
- **Instructor** at POSTECH Fall 2023
  Trustworthy ML (AIGS703L / CSED703L)

- **Teaching Assistant** at University of Pennsylvania — Spring 2015
  Machine Perception (CIS580)
- **Teaching Assistant** at University of Pennsylvania — Fall 2014
  Computer Vision and Computational Photography (CIS581)
- **Teaching Assistant** at Seoul National University — Fall 2010
  Linear Algebra for Electrical Systems
- **Instructor** at Seoul National University — Feb. 2008
  1st Free Computer Education for Gwanak-gu Community Youth

Talks

- **From LLMs to Agentic and Physical AI**
  POSTECH x 한국무역협회 최고경영자과정 — Sep. 2026
- **Towards Agentic AI Alignment**
  데이터보안프라이버시연구회 — Jun. 2026
- **AI Red Teaming: Towards AI Alignment**
  KCSE (한국소프트웨어공학학술대회) — Feb. 2026
- **AI Alignment**
  Nxt AI — Nov. 2025
  KAIST GSIS — Nov. 2025
- **AI Red Teaming and Agentic AI Security**
  OSSTEM — Nov. 2025
  Samsung SDS — Nov. 2025
  WITHCON — Nov. 2025
- **AI System for Discovering and Patching Vulnerable Code by Team Atlanta**
  POSCO DX — Oct. 2025
- **Introduction to Trustworthy LLMs**
  KISC (한국통신학회 단기강좌) — Jul. 2025
- **Trustworthy LLMs**
  Best of the Best (BoB) — Jul. 2025
- **Selective Generation Towards Trustworthy LLMs**
  NYU Frontier Lab — Jul. 2025
  KIISC AI Security Group (정보보호학회 AI보안 연구회) — Jul. 2025
- **Toward Trustworthy Large Language Models**
  PKNU AI — Dec. 2024
  POSTECH AI Day — Dec. 2024
- **Rethinking and Harnessing Trustworthiness of Generative AI for Security**
  Samsung Security Tech Forum (SSTF) – *Invited Talk* — Sep. 2024
- **Trustworthy Military-AI: AI Controllability**
  REAIM — Sep. 2024
- **Trend on Trustworthy Language Models**
  National Statistics Development Forum — Sep. 2024
- **Trustworthy AI: A Compositional Perspective**
  POSTECH AI/CSE — Apr. 2024
- **Conformal Prediction for Trustworthy AI**
  Korean AI Association Winter School — Feb. 2024

- **Uncertainty Learning for Trustworthy and Secure AI**
  POSTECH AI/CSE — Mar. 2023
  KAIST EE — Apr. 2023
  SNU CSE — Apr. 2023
  Korea University CSE — Jul. 2023
  SNU IPAI — Jul. 2023
  SNU Frontier Summer School — Aug. 2023
  UNIST IB — Oct. 2023
  KAIST EE — Jan. 2024
  CAU AI — May 2024
  Korea GSS — Jun. 2024
- **PAC Prediction Sets for AI Safety**
  ICML Workshop DFUQ 2022 – *Invited Talk* — Jul. 2022
- **Uncertainty Quantification via PAC Prediction Sets**
  DGIST — Dec. 2021
- **From Verification to Learning for Defense against Adversarial Examples in Neural Networks**
  KAIST GSIS — Aug. 2018
  Hanyang University — Aug. 2018
  KIISC — Aug. 2018